



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Mid Exam Fall 2025

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 30

Duration: 1 Hour 30 Minutes

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Design DFAs that accepts the following languages:

3x3

- a) $L = \{ w \mid w \text{ starts with } 101, \text{ contains } 010 \text{ as substring and ends with } 11 \}$ where, $\Sigma = \{0,1\}$
Example : 10101101011011, 101011011
- b) $L = \{ w \mid w \text{ starts with odd number of } b \text{ and total length of the string is even} \mid \Sigma = \{a,b\},$
Example : bbba, babbaa
- c) $L = \{ p^i q^j \mid i > 1, j > 0 \text{ and } i+j \text{ is an even number} \}$ where $\Sigma = \{ p,q \}$
Example: ppqq, pppq

2. a) Design NFAs that accept the following languages:

3x3

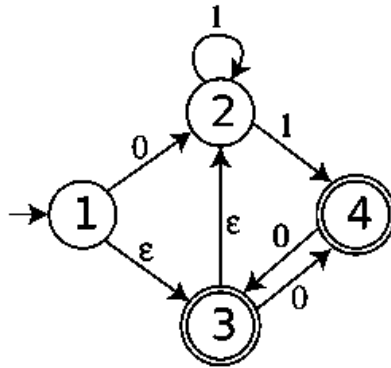
- i) $L = \{ w \mid w \text{ starts with "xyx", contains "yxz" or "xzy" and ends with "zyy" or "xxy"},$
 $\Sigma = \{ x,y,z \},$ Example : xyxxzyxxy, xyxzyzyy
- ii) $L = \{ w \mid w \text{ starts and ends with same symbol, total length of the string is a multiple of 3 if}$
the string starts with x **or** total length of the string is a **multiple of 2** if the string starts with
y or z $\}, \Sigma = \{ x,y,z \},$ Example : xyzxyx, yxzy, zyxz

b) Design ε -NFAs that accept the following language

$L = \{ w \mid w \text{ starts with "yz" or "xz", contains "zxy" as a substring, and ends with "xyx" or}$
"zyz", $\Sigma = \{ x,y,z,\varepsilon \},$ Example : yzεzxyεxyx

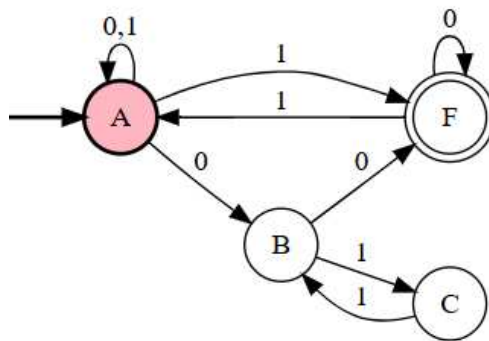
3. Consider the following NFA, and show with the help of NFA-tree whether the string “00110” is accepted or not, $\Sigma = \{0,1\}$

3



4. Convert the NFA to equivalent DFA. Show only the transition table (No state diagram needed). Here, $\Sigma = \{0,1\}$:

3



5. Design Regular Expression for the following languages

2x3

- $L = \{ w \mid w \text{ contains at least two 0's but not two consecutive 0's } \}, \Sigma = \{0,1\}$
- $L = \{ w \mid w \text{ does not contain } xxx \text{ as substring } \}, \Sigma = \{x,y\}$
- $L = \{ w \mid w \text{ starts with } pq, \text{ contains } ppq \text{ as a substring, and ends with } qpq \}, \Sigma = \{p,q\}$